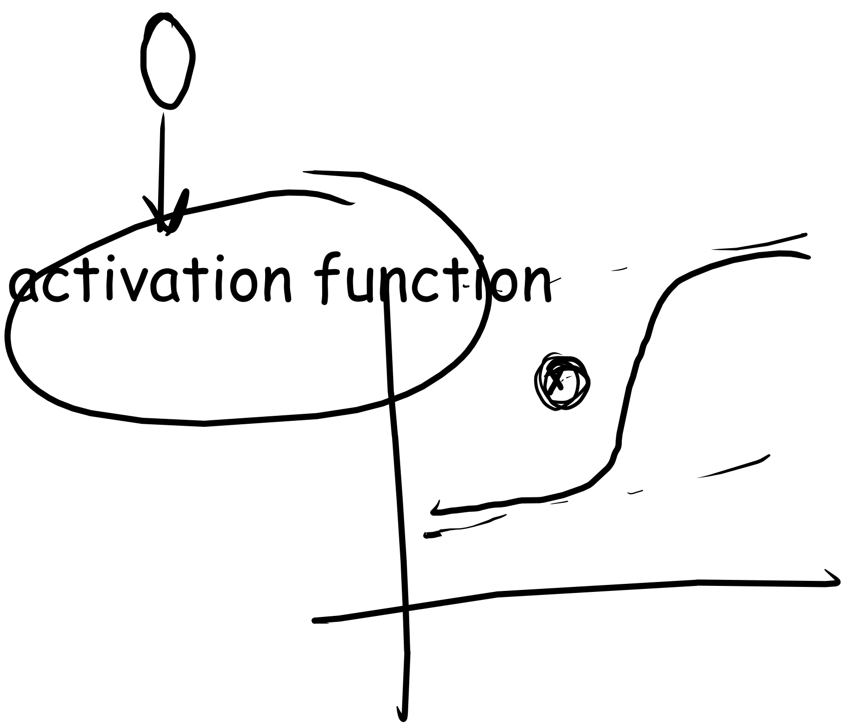
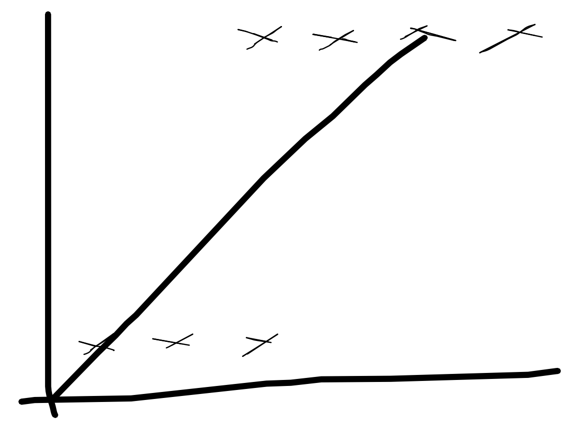
Agenda :

Activation : How strongly a neuron detect its pattern ...





linear regression

$y = mx + b$

Activation functions

① Sigmoid

Activation value will always b/w 0 & 1.

100

200

$z = w_1x_1 + w_2x_2 + w_3x_3 + \ldots w_nx_n$

Z is weighted sum

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-(w_1x_1 + w_2x_2 + w_3x_3 + \ldots + w_nx_n)}}$$

$z = 7.3$

$$\sigma(7.3) = \frac{1}{1+e^{-7.3}} = 0.93$$

$e \rightarrow 2.71$

② tanh

→ very rarely used

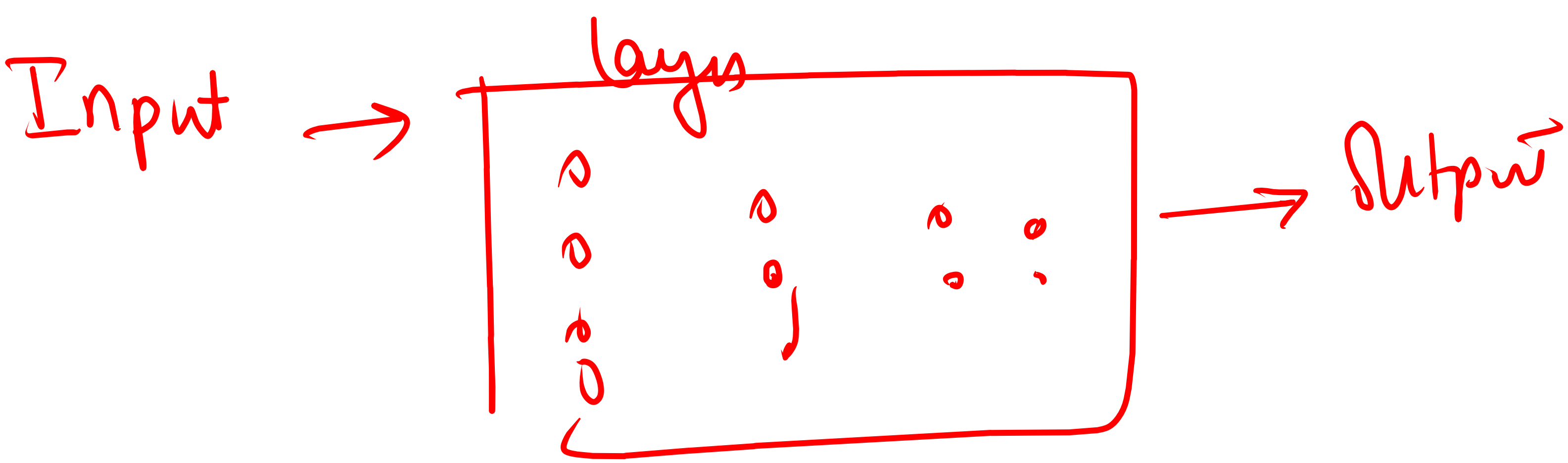
→ Now a days rather then tanh we use **ReLU**

$$\tanh = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

→ output Range = (-1 to 1)

→ Speech Recognition

→ Time series prediction.



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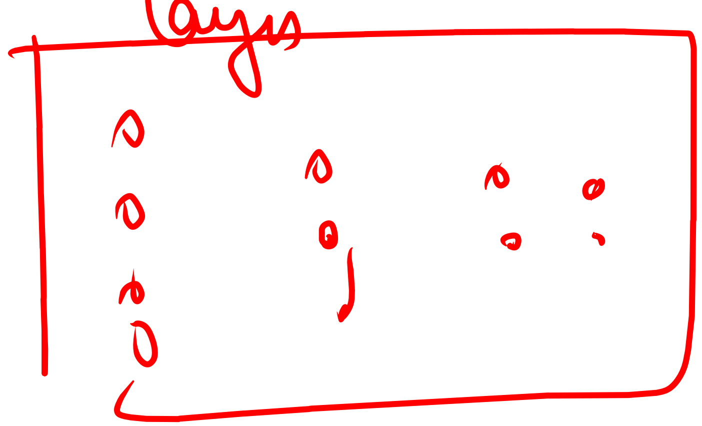
Input → layers → Output



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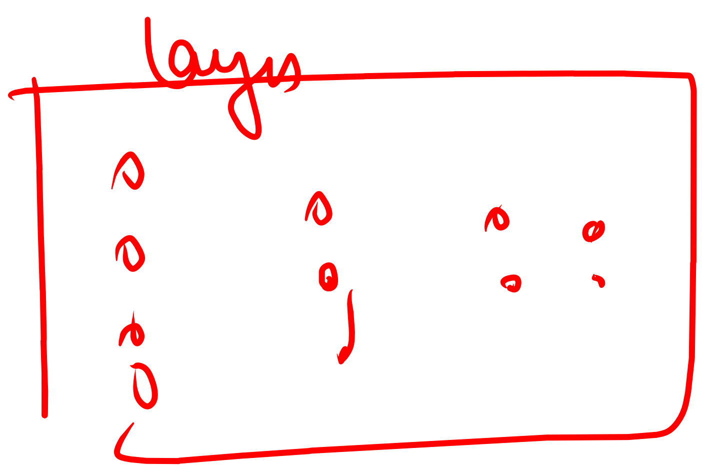
Input →



→ output

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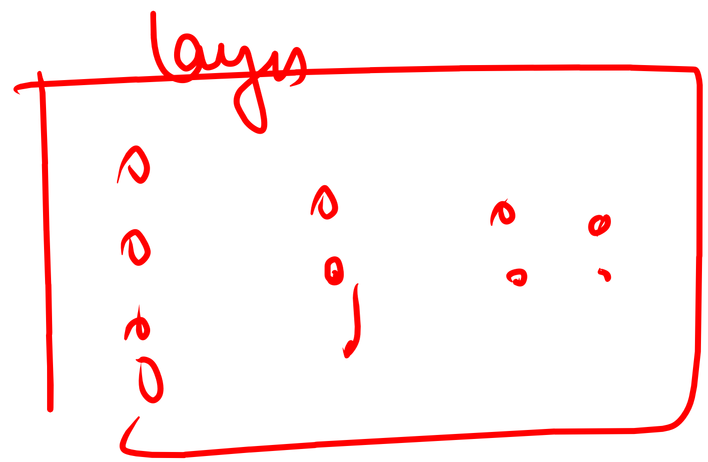
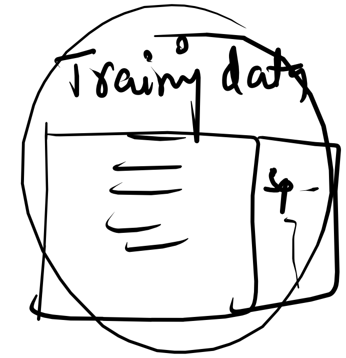


→ Output

Example2

Email is ~~spam~~ spam or non spam

$\Downarrow$ 1 (spam) — 0 (non spam)



Neuron

features	values
No. of spam words	→ 3
No. of links	→ 2
No. of attachment	→ 1

Identified by the model & neurons are trained to identify the pattern.

$x_1 = 3$

$x_2 = 2$

$x_3 = 1$

$w_1 = 0.8$

$w_2 = 0.5$

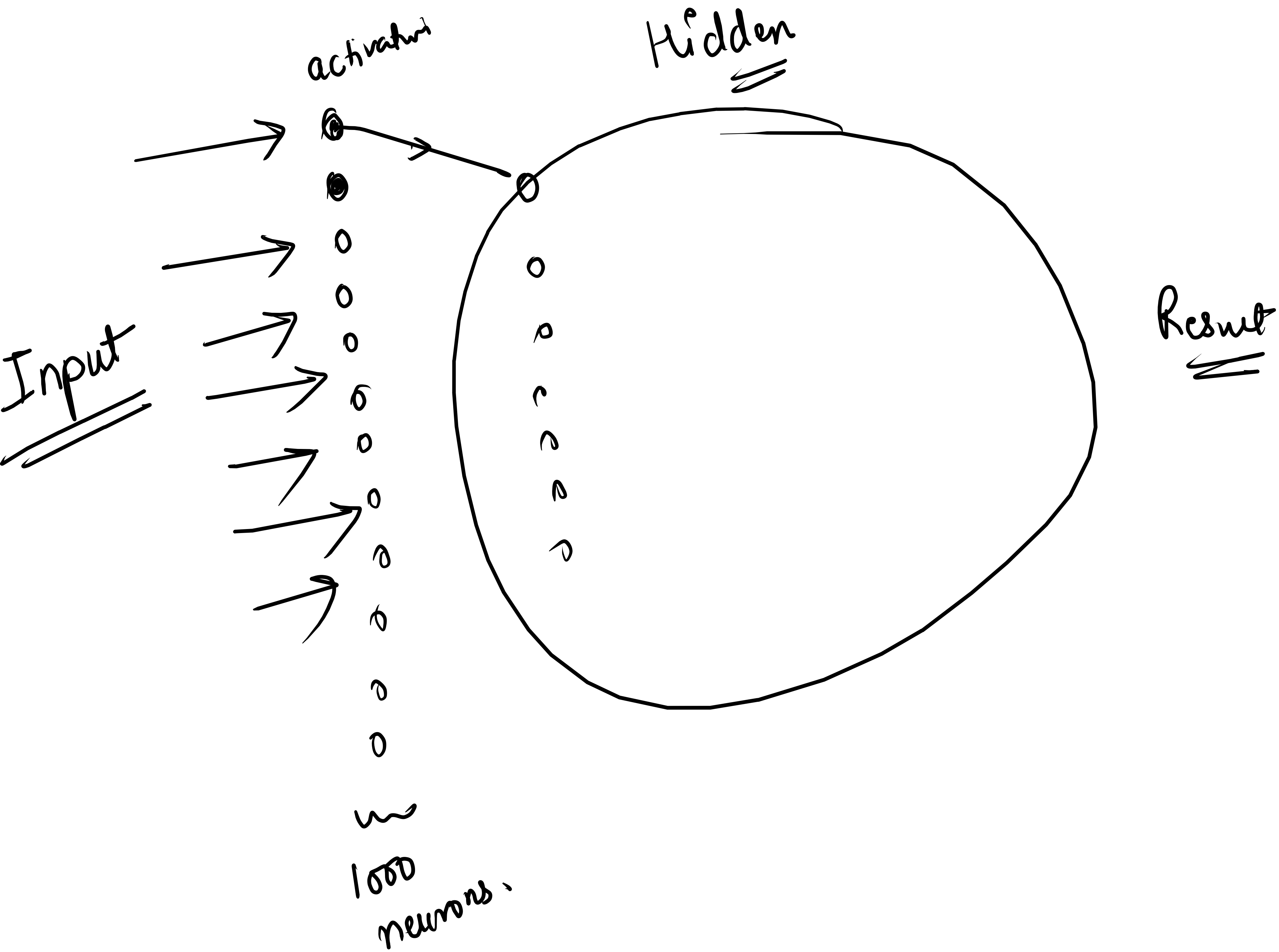
$w_3 = 0.4$

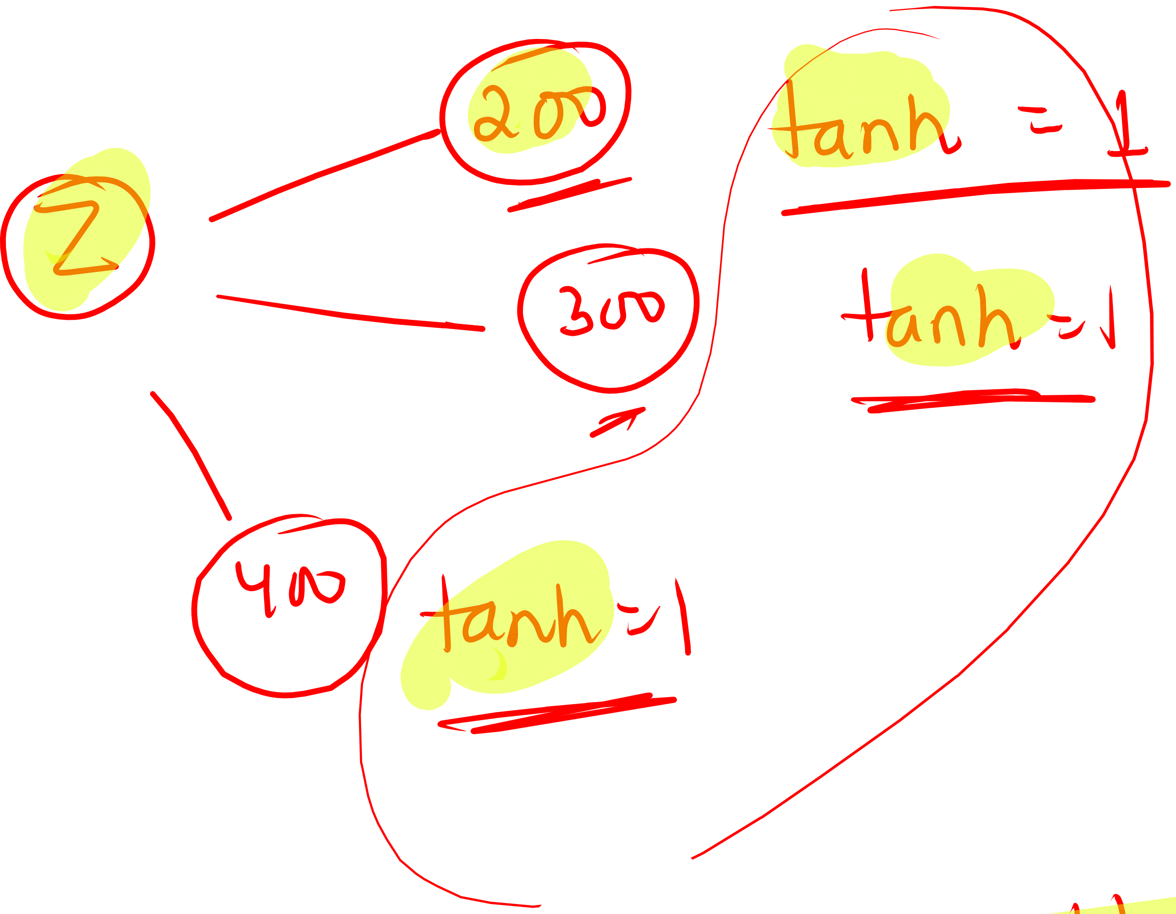
$b = -1$

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b = (0.8 \times 3) + (0.5 \times 2) + (0.4 \times 1) - 1$$

$$z = 2.8$$

Sigmoid $\Rightarrow \sigma(2.8) = \frac{1}{1+e^{-2.8}} = \frac{1}{1+\frac{1}{2.7^{(2.8)}}} = 0.91$


activation
Hidden
Input
Result
1000 neurons.



when the values become large or very small gradients become tiny making training slower

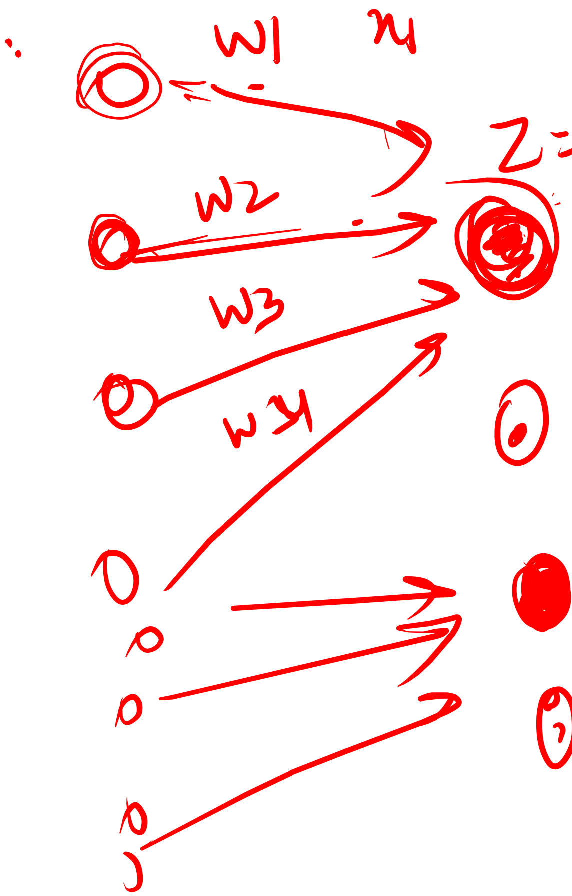
In which direction we should move the weights to reduce error (loss)

Vanishing gradient problem

ReLU

③ ReLU → Rectified linear unit.

→ $\text{ReLU}(z) = \max(0, z)$



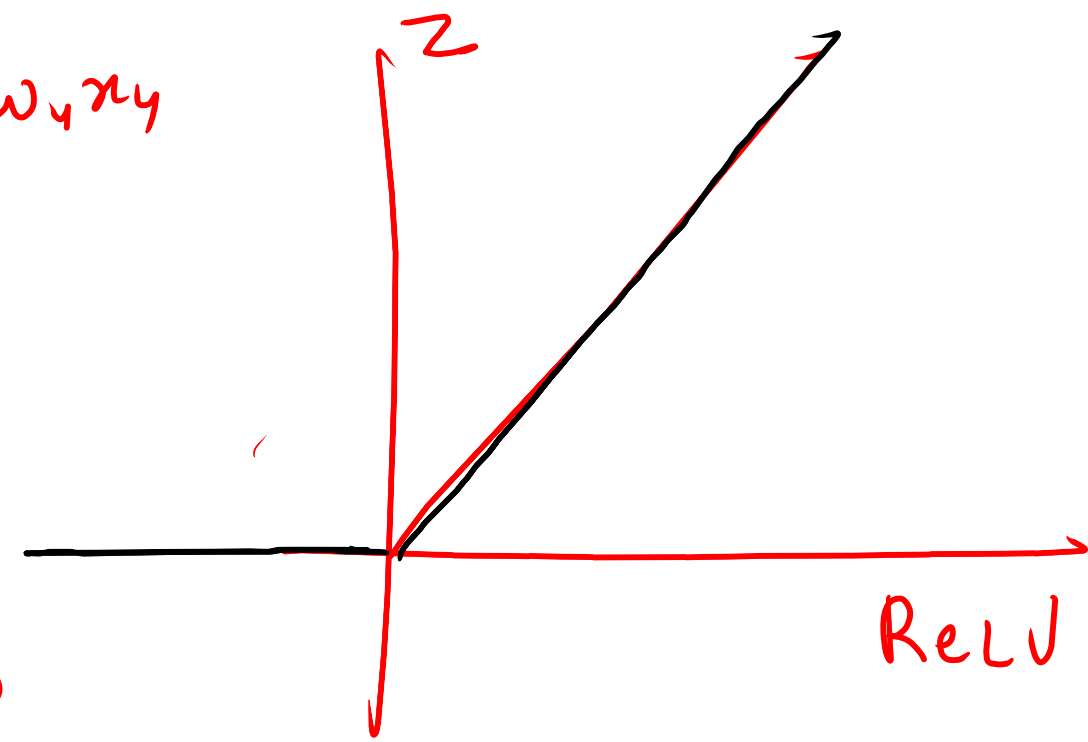
$Z = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$

$Z = 100$

$\text{ReLU}(100) = 100$

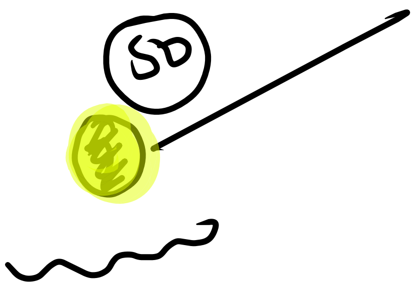
$Z = -ve = -100$

$\text{ReLU}(-100) = 0$



$Z = 50$

$\text{ReLU}(50) = 50$



make the neuron active

It should send signal to next neuron.

$Z = -100$

$\text{ReLU}(-100) = 0$

To Reduce the noise

neuron is dead.

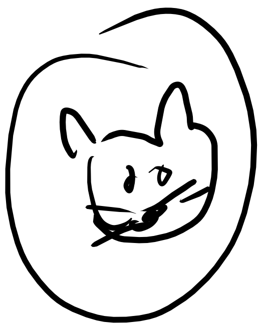
It will not send to other neuron

Types of activation function

1. Sigmoid (0-1)
2. Tanh (-1 to 1)
3. ReLU max(0, z)
4. Soft-max

Cat, Dog, Horse

Multiclass classification



probability

Cat = 0.80

Dog = 0.15

Horse = 0.05

Sum = 1

sigmoid

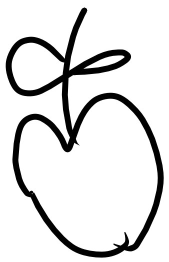
Cat = 0.8

Dog = 0.8

Horse = 0.8

Apple orange Banana pineapple grapes coconut - - - 10

0.60 0.01



Total = 1

Probability

Aman

Loss function → How wrong is my prediction is?

It compares

Actual Answer
vs
Predicted answer.

and gives error value.

Spam detection → 100 80 — spam

Spam

20 ← loss function

Calculate loss :

① Mean Square Error (MSE) :

Actual = 10

predicted = 8

$$MSE = (y - \hat{y})^2$$

$$= (10 - 8)^2$$

$$= (2)^2 = 4$$

Loss = 4 (four)

② Mean absolute error (MAE)

Actual : 10

Predicted : 8

$$MAE = |y - \hat{y}|$$

$$= |10-8|$$

$$= |2|$$

Loss = 2

③ Binary cross entropy (spam detection)

Email

Spam (1) — Notspam (0)

$y \to$ Actual $= 1$ (spam)

$p \to$ Predicted $= 0.8$

$$L = -[y \log(p) + (1-y)\log(1-p)]$$

$$= -[1 \times \log(0.8) + 0 \times \log(1-0.8)]$$

$$= -[\log(0.8) + 0]$$

$$= -[\log(0.8)] \Rightarrow 0.22$$

← small loss

0.60

0.40

0-40

Model Optimizers

Problem: Model predictions are wrong

Actual House price = 5 million
Predicted = 1 million.

loss is large

Need a mechanism to improve weights.

Optimizers → updates weights to reduce loss.

Popular Optimizers :

Gradient Descent

SGD

Momentum

RMSprop

Adam

Model Optimization Techniques

- Optimizers
- Learning Rate
- Regularization
- Normalization
- Data Augmentation
- Checkpointing
- Hyperparameter tuning.

① Optimizer → Optimizer's Job is to reduce loss.

wrong prediction

↓

Calculate error

↓

update weight

↓

Better prediction

② Learning Rate : How much should weight change ?

weight = 10

↓

9.9

↓

9.8

× very slow learning

Large → 10

↓

2

↓

-5 → 20

× Overshooting

good

10

↓

8

↓

6

↓

4

✓

③ Regularization → Model memorizes training data.

Training accuracy :- 99%.

Testing accuracy :- 50%.

} Overfitting

Dont memorize

learn pattern

4 Normalization → Inputs have very different scale.

Column

Age	Salary
25	1000000000

Normalize data

Age = 0.25

Salary → 0.70

5 Data Augmentation

Problem : Not enough data

Example : 100 cat Images

Deep learning needs 1000s of Images.

Create new Images → original cat Img

→ Rotate → flip → zoom

→ crop